

CODECOMPASS RESEARCH EVALUATION

CodeCompass Final Thesis Evaluation

Controlled bilingual retrieval, grounded QA, and Persian function documentation across three pinned Python repositories

PUBLICATION-QUALITY FINAL EVIDENCE PACKAGE

SEMANTIC HIT@3	QA USABLE	GLM PERSIAN READABILITY
94.4%	71 / 72	8.0 / 10

Frozen 2026-09-04 UTC | 36 search queries | 72 QA configurations | 18 documentation configurations

Executive Summary

CodeCompass was evaluated as a controlled research prototype for bilingual Python codebase understanding. Repository state, chunks, citations, prompts, contexts, and generation parameters were fixed; only embedding and LLM arms changed.

Gemini Embedding 2 produced the strongest semantic retrieval, while Gemini Embedding 001 produced the strongest mean downstream QA correctness. GLM outperformed Qwen under matched QA evidence. Documentation retained deterministic facts and verified citations, but Qwen documentation quality remained unavailable because of local provider execution failures.

Measured conclusion

The system met the core thesis objective within the benchmark scope. Stronger embeddings substantially improved retrieval, and a stronger LLM improved grounded answer quality and Persian rendering. The experiment does not establish a universal provider winner.

1. Experimental Design

Component	Controlled condition
Repositories	Hospital-System, CS-Bookstore, CodeCompass at pinned commits
Search	36 bilingual queries; lexical, semantic, hybrid
QA	12 questions x 3 embeddings x 2 LLMs
Documentation	9 Persian symbols x 2 LLMs
Embedding arms	Nomic local, Gemini Embedding 001, Gemini Embedding 2
LLM arms	Qwen local, GLM 5.3 Flash
Generation	temperature 0; QA max 1200; Documentation max 2400
Evaluation	randomized model-blinded human review

Repository identity

Repository	Commit	Files	Symbols	Chunks
hospital_system	da0b1f8cf04f	33	409	409
cs_bookstore	56ee148f3b4b	56	144	144
codecompass	4b9eba4df4cc	100	1236	1236

Every repository/embedding index had one vector per canonical chunk. Chunk IDs and citation metadata remained stable.

2. Search Evaluation

The lexical rows are identical across embedding arms, confirming the retrieval control. Both Gemini models materially improved semantic and hybrid retrieval.

Embedding	Method	H@1	H@3	H@5	H@10	MRR@10
Nomic local	Lexical	11.1%	33.3%	50.0%	61.1%	0.242
Nomic local	Semantic	27.8%	36.1%	38.9%	47.2%	0.328
Nomic local	Hybrid	33.3%	44.4%	47.2%	58.3%	0.406
Gemini Embedding 001	Lexical	11.1%	33.3%	50.0%	61.1%	0.242
Gemini Embedding 001	Semantic	63.9%	86.1%	94.4%	100.0%	0.771
Gemini Embedding 001	Hybrid	55.6%	77.8%	88.9%	94.4%	0.689
Gemini Embedding 2	Lexical	11.1%	33.3%	50.0%	61.1%	0.242
Gemini Embedding 2	Semantic	75.0%	94.4%	97.2%	100.0%	0.853
Gemini Embedding 2	Hybrid	47.2%	83.3%	94.4%	97.2%	0.661

Bilingual results

Lang.	Embedding	Sem. H@3	Sem. MRR	Hybrid H@3	Hybrid MRR
EN	Nomic local	72.2%	0.647	83.3%	0.741
EN	Gemini Embedding 001	88.9%	0.803	88.9%	0.819
EN	Gemini Embedding 2	94.4%	0.868	88.9%	0.743
FA	Nomic local	0.0%	0.009	5.6%	0.071
FA	Gemini Embedding 001	83.3%	0.739	66.7%	0.559
FA	Gemini Embedding 2	94.4%	0.838	77.8%	0.579

Interpretation

Gemini Embedding 2 led semantic ranking. Gemini Embedding 001 led Hybrid Hit@1 and Hybrid MRR, whereas Gemini Embedding 2 led Hybrid Hit@3, Hit@5, and Hit@10. Hybrid fusion therefore changes the relative ordering of embedding models.

3. Search Runtime

Latency is descriptive of the recorded environment. External-provider and local timings are not service-level guarantees.

Embedding	Method	Median ms	P95 ms	Range ms
Nomic local	Lexical	30.971	203.125	9.318-207.47
Nomic local	Semantic	11.317	17.96	5.803-21.799
Nomic local	Hybrid	36.883	203.3	17.114-248.693
Gemini Embedding 001	Lexical	34.258	200.899	11.075-251.178
Gemini Embedding 001	Semantic	12.607	23.445	6.409-27.257
Gemini Embedding 001	Hybrid	36.281	211.891	17.083-234.098
Gemini Embedding 2	Lexical	29.365	193.118	9.996-254.064
Gemini Embedding 2	Semantic	11.503	22.982	6.59-25.251
Gemini Embedding 2	Hybrid	43.182	186.681	18.71-194.506

4. Grounded QA

Execution reliability

Model	Initial	Retry 1	Retry 2	Failed	Total
Qwen local	27	9	0	0	36
GLM 5.3 Flash	24	0	11	1	36
Overall	51	9	11	1	72

Human quality by LLM

LLM	n	Correct.	Grounded	FA read.	Useful
GLM 5.3 Flash	35	7.943	8.314	8.000	8.286
Qwen local	36	6.972	7.000	5.333	6.250

Paired finding

Across 35 matched outputs with identical question and embedding evidence, GLM minus Qwen was +1.000 correctness, +1.343 groundedness, +2.824 Persian readability, and +2.086 usefulness. Seven outputs had provider-confirmed length termination; one GLM combination remained unavailable.

5. QA Embedding and Runtime Effects

Embedding	n	Correct.	Grounded	FA read.	Useful
Gemini Embedding 001	23	7.783	7.913	6.545	7.435
Gemini Embedding 2	24	7.000	7.333	6.667	7.000
Nomic local	24	7.583	7.708	6.667	7.333

A stronger retriever did not monotonically improve generated answers. Gemini 001 had the highest mean QA correctness, while Gemini 2 had the highest semantic retrieval scores. This is evidence that retrieval and generation are distinct constraints.

LLM	n	Mean s	Median s	P95 s	Range s
Qwen local	36	64.688	67.172	100.152	20.002-103.705
GLM 5.3 Flash	35	122.156	10.753	75.939	5.039-3796.914

Token usage was not measured because provider responses did not expose usage fields in the saved artifacts.

6. Function Documentation

Documentation combines deterministic syntax facts and citations with model-generated prose. Structural identity is not delegated to the LLM.

LLM	Usable	Unavailable	Correct.	Grounded	FA read.	Useful
GLM 5.3 Flash	9	0	6.778	8.222	8.111	8.111
Qwen local	0	9	Not measured	Not measured	Not measured	Not measured

All nine GLM cases completed normally and citation identity mismatches were zero. All Qwen attempts and the separately preserved recovery attempts failed with `provider_failure/http_error`. The Qwen cells are unavailable measurements, not zero-quality scores.

7. Error and Hallucination Analysis

Human label	Count	Publication interpretation
No hallucination	73	Preserved from the blinded review
No hallucination, but vague	1	Preserved from the blinded review
Explicit hallucination	4	Preserved from the blinded review
Mild or implicit	2	Preserved from the blinded review

Failures were not hidden or imputed. The final dataset retains one GLM empty-content QA failure, nine Qwen Documentation provider failures, seven confirmed length terminations, and additional outputs that appeared incomplete without a length signal. Hallucination labels remain reviewer judgments and are not inferred from generation success.

8. Scientific Findings

1. Embedding capability materially affected retrieval, especially semantic search and Persian alignment.
2. Retrieval gains did not guarantee monotonic downstream QA gains; context composition and generation capability remained important.
3. GLM produced stronger matched QA quality and substantially better Persian readability than local Qwen.
4. Deterministic documentation facts preserved source identity and citation reliability even when provider execution failed.
5. CodeCompass met the measured thesis objective as a bilingual, evidence-grounded Python codebase assistant, while remaining a research prototype rather than a universally validated production system.

Limitations

Limitation	Consequence
Three Python repositories	Generalization is unmeasured.
12 base QA questions	Results are paired and descriptive.
One human reviewer	No inter-rater reliability estimate.
Missing executions	No score imputation; comparisons have unequal n.
External providers	Privacy, cost, availability, and latency differ.
Token-limited outputs	Some answers are incomplete at max_tokens=1200.

9. Final Conclusion

The final evidence supports CodeCompass as a successful thesis prototype within its declared scope. Stronger embeddings improved retrieval, a stronger LLM improved grounded answer quality and Persian prose, and deterministic metadata preserved citations and documentation facts. The most defensible deployment interpretation is configurable: local models preserve offline privacy, while external models provide the strongest measured quality. Remaining negative results are explicit and reproducible.

10. Reproducibility

Artifact	SHA-256
benchmark_cases.json	ec134348d3b0cb24e062b2d663a4521b5630dc449f4bb27e3bb1461a3536974f
freeze_manifest.json	3032f88c17bd23eae170a4a97dfac385a46868b666d322ae088a33bfe20f5c67
retrieval_results.json	2e5c36bfd9ed581674f4705fd3498d8e3d234f40bc1239a64fab7e39c191fa56
qa_results.json	e96935f5d3903a734cec9e52dbfeac32bf3e60489244e6991b584246a1a12fd1
qa_recovery_results.json	3e5bd12d1d4b87c5b5f8d05271b7e898b7ec9f97e45bffe3aacefa505c5fcfab
qa_glm_second_recovery_results.json	3d26f8b9d63b227cd7a2ca4bfe2e696ee9937e1623f568a9dbab500b3fabdf60
documentation_results.json	85b4fc6b448177e8cfb9771b94deaec8db4f27fa0328439893359d1620b90d90
documentation_qwen_recovery_results.json	40fba2049fd386f69aa8414e7e0c3f334876d59b5165bada60d2acdd33405981
human_review_blinded.json	715d64dc60f4f5e53b5768d12ee22d1ffb7ca93dac2a9748c555028f256cbbaf
blind_mapping.json	cd2b3c09ad9ece447a1ddf923acb9f29457bec0df13af8eb1b1973b1c24fd225
human_review_scored.json	4dbfb5b38259aad4bd189c3e5bfc18186718e6286ff48654874a94334057a97b
human_evaluation_summary.json	5e447f000df58a4948219d85fd41e95a57b8bc4c978242a1448f826c7732ff9b

Reporting made zero provider, retrieval, embedding, or indexing calls. No scores or raw experiment records were modified.

Appendix A. Case-Level Human Scores

Unavailable executions remain unscored. FA readability is measured only for Persian outputs.

Blind ID	Case	Task	Repo	Lang	Embedding	LLM	Status	C	G	FA	U
R-001	FTE-DOC-B-03	documentati on	cs_bookstore	fa	-	glm	complete	9	10	9	9
R-002	FTE-QA-C-EN-02	qa	codecompass	en	gemini_2	glm	complete	9	9	-	9
R-003	FTE-QA-C-EN-01	qa	codecompass	en	gemini_001	qwen	complete	6	7	-	4
R-004	FTE-QA-B-FA-02	qa	cs_bookstore	fa	nomic	glm	complete	8	9	8	8
R-005	FTE-QA-C-EN-01	qa	codecompass	en	gemini_001	glm	complete	9	9	-	9
R-006	FTE-QA-H-EN-01	qa	hospital_system	en	gemini_001	qwen	complete	8	8	-	8
R-007	FTE-QA-C-EN-02	qa	codecompass	en	nomic	glm	complete	9	9	-	9
R-008	FTE-QA-C-EN-01	qa	codecompass	en	gemini_2	glm	complete	8	8	-	8
R-009	FTE-QA-C-FA-01	qa	codecompass	fa	gemini_001	glm	complete	6	8	8	8
R-010	FTE-QA-B-FA-01	qa	cs_bookstore	fa	gemini_001	glm	complete	8	8	8	8
R-011	FTE-QA-B-FA-01	qa	cs_bookstore	fa	gemini_2	glm	complete	8	8	8	8
R-012	FTE-QA-H-FA-02	qa	hospital_system	fa	gemini_2	qwen	complete	8	8	8	8
R-013	FTE-QA-B-EN-02	qa	cs_bookstore	en	gemini_001	qwen	complete	1	1	-	1
R-014	FTE-QA-C-FA-01	qa	codecompass	fa	gemini_2	glm	complete	6	8	8	8
R-015	FTE-QA-H-FA-01	qa	hospital_system	fa	gemini_001	qwen	complete	8	8	0	4
R-016	FTE-QA-C-EN-01	qa	codecompass	en	nomic	qwen	complete	8	8	-	8
R-017	FTE-QA-C-EN-02	qa	codecompass	en	nomic	qwen	complete	3	3	-	3
R-018	FTE-DOC-H-02	documentati on	hospital_system	fa	-	qwen	failed	-	-	-	-
R-019	FTE-QA-C-FA-02	qa	codecompass	fa	nomic	glm	complete	6	8	8	8
R-020	FTE-QA-H-FA-02	qa	hospital_system	fa	nomic	qwen	complete	8	8	8	8
R-021	FTE-DOC-B-03	documentati on	cs_bookstore	fa	-	qwen	failed	-	-	-	-
R-022	FTE-QA-C-EN-02	qa	codecompass	en	gemini_001	glm	complete	9	9	-	9
R-023	FTE-QA-H-EN-02	qa	hospital_system	en	gemini_001	glm	complete	9	9	-	9
R-024	FTE-QA-B-FA-02	qa	cs_bookstore	fa	gemini_2	qwen	complete	8	8	8	8
R-025	FTE-QA-C-FA-01	qa	codecompass	fa	nomic	qwen	complete	8	8	0	4
R-026	FTE-QA-H-FA-01	qa	hospital_system	fa	gemini_2	qwen	complete	8	8	0	4
R-027	FTE-QA-B-FA-02	qa	cs_bookstore	fa	gemini_001	qwen	complete	8	8	8	8
R-028	FTE-QA-H-EN-01	qa	hospital_system	en	nomic	glm	complete	8	8	-	8
R-029	FTE-QA-B-EN-01	qa	cs_bookstore	en	gemini_001	qwen	complete	8	8	-	8
R-030	FTE-QA-H-FA-02	qa	hospital_system	fa	gemini_2	glm	complete	8	8	8	8
R-031	FTE-QA-B-FA-01	qa	cs_bookstore	fa	gemini_2	qwen	complete	8	8	8	8
R-032	FTE-QA-C-FA-01	qa	codecompass	fa	gemini_2	qwen	complete	8	8	0	4
R-033	FTE-QA-H-FA-02	qa	hospital_system	fa	nomic	glm	complete	8	8	8	8
R-034	FTE-QA-B-FA-02	qa	cs_bookstore	fa	gemini_2	glm	complete	6	8	8	8
R-035	FTE-QA-B-FA-02	qa	cs_bookstore	fa	gemini_001	glm	complete	8	8	8	8
R-036	FTE-QA-B-EN-01	qa	cs_bookstore	en	gemini_2	qwen	complete	8	8	-	8
R-037	FTE-QA-B-EN-01	qa	cs_bookstore	en	nomic	qwen	complete	8	8	-	8
R-038	FTE-QA-H-EN-02	qa	hospital_system	en	gemini_2	glm	complete	9	9	-	9

Blind ID	Case	Task	Repo	Lang	Embedding	LLM	Status	C	G	FA	U
R-039	FTE-QA-B-EN-01	qa	cs_bookstore	en	nomic	glm	complete	8	8	-	8
R-040	FTE-DOC-B-02	documentati on	cs_bookstore	fa	-	glm	complete	6	8	8	8
R-041	FTE-QA-C-EN-02	qa	codecompass	en	gemini_001	qwen	complete	9	9	-	9
R-042	FTE-DOC-C-01	documentati on	codecompass	fa	-	glm	complete	6	8	8	8
R-043	FTE-QA-C-EN-01	qa	codecompass	en	gemini_2	qwen	complete	8	8	-	8
R-044	FTE-DOC-H-01	documentati on	hospital_system	fa	-	glm	complete	6	8	8	8
R-045	FTE-QA-B-EN-02	qa	cs_bookstore	en	nomic	qwen	complete	1	1	-	1
R-046	FTE-QA-B-EN-02	qa	cs_bookstore	en	gemini_001	glm	complete	9	9	-	9
R-047	FTE-QA-C-FA-01	qa	codecompass	fa	nomic	glm	complete	8	8	8	8
R-048	FTE-DOC-H-02	documentati on	hospital_system	fa	-	glm	complete	8	8	8	8
R-049	FTE-QA-C-FA-02	qa	codecompass	fa	gemini_2	qwen	complete	8	8	8	8
R-050	FTE-QA-H-EN-02	qa	hospital_system	en	gemini_2	qwen	complete	1	1	-	1
R-051	FTE-QA-B-EN-01	qa	cs_bookstore	en	gemini_2	glm	complete	6	8	-	8
R-052	FTE-QA-B-EN-02	qa	cs_bookstore	en	nomic	glm	complete	9	9	-	9
R-053	FTE-QA-C-FA-02	qa	codecompass	fa	nomic	qwen	complete	8	8	8	8
R-054	FTE-QA-H-FA-02	qa	hospital_system	fa	gemini_001	glm	failed	-	-	-	-
R-055	FTE-DOC-C-01	documentati on	codecompass	fa	-	qwen	failed	-	-	-	-
R-056	FTE-QA-H-FA-01	qa	hospital_system	fa	gemini_2	glm	complete	6	8	8	8
R-057	FTE-QA-C-EN-02	qa	codecompass	en	gemini_2	qwen	complete	3	3	-	3
R-058	FTE-QA-H-FA-01	qa	hospital_system	fa	gemini_001	glm	complete	8	8	8	8
R-059	FTE-DOC-C-03	documentati on	codecompass	fa	-	glm	complete	6	8	8	8
R-060	FTE-QA-B-EN-01	qa	cs_bookstore	en	gemini_001	glm	complete	8	8	-	8
R-061	FTE-DOC-H-01	documentati on	hospital_system	fa	-	qwen	failed	-	-	-	-
R-062	FTE-QA-H-EN-01	qa	hospital_system	en	gemini_2	qwen	complete	8	8	-	8
R-063	FTE-QA-H-FA-01	qa	hospital_system	fa	nomic	glm	complete	8	8	8	8
R-064	FTE-QA-C-FA-02	qa	codecompass	fa	gemini_001	glm	complete	8	8	8	8
R-065	FTE-DOC-H-03	documentati on	hospital_system	fa	-	glm	complete	8	8	8	8
R-066	FTE-DOC-C-02	documentati on	codecompass	fa	-	qwen	failed	-	-	-	-
R-067	FTE-QA-H-EN-01	qa	hospital_system	en	nomic	qwen	complete	8	8	-	8
R-068	FTE-QA-B-FA-01	qa	cs_bookstore	fa	gemini_001	qwen	complete	8	8	8	8
R-069	FTE-QA-C-FA-02	qa	codecompass	fa	gemini_001	qwen	complete	8	8	8	8
R-070	FTE-QA-H-EN-02	qa	hospital_system	en	nomic	qwen	complete	9	9	-	9
R-071	FTE-DOC-H-03	documentati on	hospital_system	fa	-	qwen	failed	-	-	-	-
R-072	FTE-DOC-C-02	documentati on	codecompass	fa	-	glm	complete	6	8	8	8
R-073	FTE-QA-H-FA-02	qa	hospital_system	fa	gemini_001	qwen	complete	8	8	8	8
R-074	FTE-QA-H-EN-01	qa	hospital_system	en	gemini_2	glm	complete	8	8	-	8
R-075	FTE-QA-B-FA-02	qa	cs_bookstore	fa	nomic	qwen	complete	8	8	8	8

Blind ID	Case	Task	Repo	Lang	Embedding	LLM	Status	C	G	FA	U
R-076	FTE-QA-H-EN-02	qa	hospital_system	en	nomic	glm	complete	9	9	-	9
R-077	FTE-QA-B-FA-01	qa	cs_bookstore	fa	nomic	qwen	complete	8	8	8	8
R-078	FTE-QA-C-FA-02	qa	codecompass	fa	gemini_2	glm	complete	8	8	8	8
R-079	FTE-QA-H-EN-01	qa	hospital_system	en	gemini_001	glm	complete	8	8	-	8
R-080	FTE-QA-B-EN-02	qa	cs_bookstore	en	gemini_2	qwen	complete	1	1	-	1
R-081	FTE-QA-B-FA-01	qa	cs_bookstore	fa	nomic	glm	complete	8	8	8	8
R-082	FTE-QA-C-FA-01	qa	codecompass	fa	gemini_001	qwen	complete	8	8	0	4
R-083	FTE-DOC-B-01	documentati on	cs_bookstore	fa	-	glm	complete	6	8	8	8
R-084	FTE-QA-H-EN-02	qa	hospital_system	en	gemini_001	qwen	complete	9	9	-	9
R-085	FTE-QA-C-EN-01	qa	codecompass	en	nomic	glm	complete	8	8	-	8
R-086	FTE-DOC-B-01	documentati on	cs_bookstore	fa	-	qwen	failed	-	-	-	-
R-087	FTE-DOC-B-02	documentati on	cs_bookstore	fa	-	qwen	failed	-	-	-	-
R-088	FTE-QA-B-EN-02	qa	cs_bookstore	en	gemini_2	glm	complete	9	9	-	9
R-089	FTE-QA-H-FA-01	qa	hospital_system	fa	nomic	qwen	complete	8	8	0	4
R-090	FTE-DOC-C-03	documentati on	codecompass	fa	-	qwen	failed	-	-	-	-